A Vision-Based Sign Language Recognition & Voice Conversion System for Inclusive Communication

Problem Addressed

Sign language is an important communication method for deaf and speech-impaired individuals. However, most people do not understand sign language, which creates a communication barrier between deaf individuals and the general public. Therefore, a system that can automatically recognize sign language gestures and convert them into text is needed to improve communication.

Research Gap

Many existing sign language recognition systems require special hardware such as data gloves or sensors. These systems can be expensive and uncomfortable for users. In addition, some camera-based systems suffer from low accuracy due to lighting conditions and background noise.

Key Contribution

The paper proposes a **real-time sign language recognition system using MediaPipe hand tracking and machine learning algorithms**. The system detects hand landmarks from video input and uses these features to classify sign language gestures.

Methodology

- **Data Acquisition**

The system captures video using a camera to detect hand gestures in real time.

- **Hand Detection**

The **MediaPipe framework** is used to detect hand landmarks. It identifies **21 key points of the hand**, including finger joints and palm position.

- **Feature Extraction**

The coordinates of the detected landmarks are extracted and used as features representing the hand gesture.

- **Gesture Classification**

Machine learning algorithms are used to classify the gesture based on the extracted features. The system predicts the corresponding sign language character or gesture.

Results

The system demonstrates effective real-time gesture recognition with high accuracy. MediaPipe hand tracking improves detection performance by accurately identifying hand landmarks and reducing background noise effects.

Advantages

The proposed system provides several benefits:

- Real-time gesture recognition
- Low-cost implementation using a standard camera
- No need for wearable devices
- Efficient and accurate hand landmark detection

Conclusion

The research shows that combining **MediaPipe hand tracking with machine learning techniques** can successfully recognize sign language gestures in real time. This approach provides a practical solution for improving communication between deaf individuals and others.